

Distributed System Operating s/w

To make a success of parallel processing one has to ensure that the overall computational load is distributed as evenly as possible across the available processors. One possible approach to this problem is to follow the well established practice of single processor programming, and shield the parallel programmer from the complexity of the underlying hardware by an operating system (OS). Taos is a distributed OS that implements this approach. It supports shrink wrapped binary portable/parallel s/w running on a network of mixed, heterogeneous processors of arbitrary configuration.

Taos applications, in C or assembly, are written for a Virtual Processor or VP. The VP has all the attributes of a real processor, except that it is not available in SI. This 32-bit machine has 16 registers, and supports standard data types and addressing modes. Applications are compiled (for C programs) or assembled into the machine code of the VP. At run time, when the code is loaded into a processor for execution, it is translated from VP code into that of the host processor. Currently translators for the T800 and TMS320C40 are available.

Taos programs comprise two basic types of objects. Tool objects correspond loosely to conventional programming procedures, while control objects bind the tools into complete programs. Program execution begins by assigning the program's main control object to a convenient processor. The OS uses the control

object to define the hardware resources required by the program and to identify the main tool object of the program. This tool object is then brought into the processor and translated into the processor's machine code. The program execution proceeds as the main tool, in turn, calls subsidiary tools. Execution continues as a single thread of sequential code within this processor until either the program terminates, or a tool calls a further control object, initiating a

effect, applications are created on the fly. This ensures highly efficient use of the processor's computational and memory resources. Also, objects can be designated as either non-virtual or virtual. Non-virtual objects are brought into the host processor at the earliest opportunity, remaining there until the execution thread terminates. Virtual objects are brought in at the last moment, and deleted almost immediately after use. Another memory saving

expand. Under Taos the network is self balancing. Running on every processor is a copy of a Taos micro-kernel. This provides all the standard functions common to any OS kernel together with a number of functions specific to Taos, including distributed load balancing.

One Taos application, now available, is a ray tracing package based on a standard PC and T800 plug-in card. Supporting a wide range of standard features such as importing 3D objects from standard CAD packages, a comprehensive set of graphics primitives, plus any number of light sources of any colour, the s/w can create a photorealistic image in minutes as opposed to tens of hours with existing systems. In theory, a ray tracer based on an array of 50 C40's should be able to generate video images in real-time. Taos is not, however, limited to ray-tracing. It has been designed for use on networked systems from embedded controllers through the multi-user, multi-tasking UNIX box to the massively parallel supercomputer. The basic requirement for running s/w under Taos is to develop a set of Taos objects that implement a set of services on a network according to a well defined application-programmer interface (API). Objects for API's compliant with leading operating systems, especially variants of UNIX, will be developed during 1992.

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This 740x578 pixel, 24-bit colour image containing over 500 objects was generated on an array of eight transputers in under 11min.

second sequential thread of computation. This second thread can either run on the original processor, or migrate to another for parallel execution.

With Taos, objects are held as unbound, independent entities on disk, and are not linked into the thread of execution until they are brought into the host processor. In

technique is the sharing of objects between execution threads whenever possible.

The use of a master program to track the current load on the network and place new work in a way that keeps the network balanced requires high overheads in large networks and constrains the ability of the network to

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